

Anokh Surpur

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SUMMARY

AI Engineering undergraduate at KLE Technological University with proven experience designing and shipping distributed AI systems from scratch. Skilled in Python, PyTorch, federated learning, blockchain, FastAPI, and data analytics, with hands-on experience across the full stack, from training neural networks and building REST APIs to deploying real-time dashboards and blockchain ledgers across physical machines. Applies data-driven thinking, structured problem solving, and a bias for execution to engineering and technology roles.

INTERNSHIP EXPERIENCE

Research Intern, Centre for AI Research (CAIR), KLE Technological University

Jan 2025 - May 2025

Project: FedMedChain, Reputation-Weighted Federated Learning with Blockchain Incentives. Mentor: Prof. S. Chickerur, KLE Tech

- Deployed a working three-machine federated learning system across a real LAN (Linux server and 2 Windows hospital nodes) where each hospital trains a ResNet-18 model locally and submits only weights, so no raw patient data ever leaves the machine, achieving 96% global model accuracy and a verified 3:1 contribution ratio in Round 1.
- Built a SHA-256 blockchain ledger and SQLite reputation database that immutably records every federated round's accuracy metrics, dataset hash commitments, reputation changes, and MedCoin reward allocations, persisting across server restarts.
- Shipped a live HTML/JS analytics dashboard with real-time FastAPI endpoints displaying reputation scores, blockchain state, and round-by-round model performance, plus a cosine-similarity poisoning detector that flags malicious weight submissions before aggregation.

KEY PROJECTS

Agentic Rumor Simulation, LLM-Driven Social Dynamics

Python, LangChain, NetworkX, Bayesian Inference

- Built LLM-powered autonomous agents that simulate how rumors spread, mutate, and die across social networks, with each agent holding a dynamic belief vector updated via Bayesian inference after every interaction to mimic how real people revise opinions based on trust and source credibility.
- Constructed large-scale social graphs with NetworkX (Girvan-Newman and LFR benchmark models) to replicate real-world community clustering and trust-weighted connections in Indian social contexts.
- Ran controlled experiments with Independent Cascade and Linear Threshold diffusion models, injecting fact-check interventions at timed steps, and logged belief entropy, rumor half-life, and intervention efficiency.

Yet Another Game-N-Gen, Self-Supervised Multi-Game Simulator

PyTorch, Stable Diffusion, DQN, Hugging Face

- Built a generative AI system that learns to simulate Chess, Snake, and other games purely from video frames, with no hand-coded rules, by reformulating game simulation as a self-supervised latent prediction task using a modified Stable Diffusion v1.5 conditioned on prior frames and action tokens, achieving 29.8 dB PSNR on next-frame prediction.
- Built and open-sourced a 670 GB+ dataset generated by DQN agents, handling custom tokenization, domain-adapted VAE decoder fine-tuning, and large-scale training on DGX H100 GPUs.
- Released pretrained models on Hugging Face as one unified model that handles multiple games, eliminating the need for separate simulators per game.

Mobile-Friendly MaxViT, Edge AI Skin-Lesion Classifier on iOS

PyTorch, CoreML, NAS, Knowledge Distillation

- Developed the first MaxViT vision transformer compatible with the Once-for-All (OFA) NAS framework for on-device medical AI on iPhones.
- Trained a MaxViT-Small teacher model on HAM10000, then distilled into compact student models, achieving 82.74% macro F1 with a 49.43% size reduction and 66.7% drop in compute cost.
- Deployed student models to iOS via CoreML for real-time on-device skin-lesion classification with zero cloud dependency.
- Open-sourced the complete PyTorch pipeline on GitHub, covering data preprocessing, NAS architecture search, and distillation.

EDUCATION

B.E. Computer Science (Artificial Intelligence), KLE Technological University, Hubballi

2021 - 2026

Pre-University Course (PUC), Science, Prism PU Science College, Dharwad

2019 - 2021

TECHNICAL SKILLS

Languages: Python, C, SQL, JavaScript, HTML/CSS

AI / ML: PyTorch, TensorFlow, MONAI, Flower (flwr), Stable Diffusion, LangChain, NetworkX

Data & Analytics: pandas, NumPy, scikit-learn, SQLite, FastAPI, REST APIs

Infrastructure & Tools: Docker, SHA-256 Blockchain, IPFS, Hyperledger Fabric, Git, Hugging Face, CoreML, Android Studio